

8 The interface and instrumentality of Eurorack modular synthesis

Eliot Bates

The performance of synthesizers has had a transformative effect on the sound of music of nearly all kinds, from Arab-American ensembles in Detroit (Rasmussen 1996) to mbaqanga in South Africa (Meintjes 2003), in addition to being constitutive of hundreds of new transnational and translocal genres and subgenres of electronic music such as house, techno, electro, and hip-hop (Eshun 1998). The designing and making of synthesizers, correspondingly, has transformed global and national industries for the making and marketing of musical instruments (Théberge 1997) and audio technologies more generally. Synthesizers are not morphologically just one *kind* of thing, as they can take the form of a keyboard instrument (Pinch and Trocco 2002), a drum machine (Butler 2014), a guitar pedal (Bonanno 1984), a portable video game system, a big rack of stuff that looks like test or weapons control equipment, or software “virtual” instruments and smartphone apps (D’Errico 2016). They are not even obligatorily concerned with synthesizing *sound* and can be used for synthesizing video (Sen 2019).

This multiplicity of morphological forms and flexibility in terms of musical (or other) ends are two of the many aspects of synthesizers that make them especially interesting for study, whether the goal is a more “organological” consideration of the properties of musical instruments and their construction (Bates 2021) or a more technological-sociological consideration of synthesizers as objects that become meaningful within and transform culture/society. Perhaps because of their obvious dual nature as technologies *and* instruments, synthesizers have interested science and technology studies (STS) scholars. Indeed, Trevor Pinch and Frank Trocco’s (2002) history of the Moog synthesizer is one of the best-known book-length expositions of the social construction of technology (SCOT) field. Related research based on “following the users” influenced the STS subfield of user studies (Oudshoorn and Pinch 2003) and the trajectory of sound studies that emerged from STS (Pinch and Bijsterveld 2004).

In this essay, I will examine a couple of aspects of hardware modular synthesizers which, despite not being a “new” morphological form (most commercially sold synthesizers from the 1960s through the early 1970s were modular), experienced an extremely unlikely expansion of interest and activity in the mid-1990s. With a contemporary modular synth, users buy a powered rack that holds modules specific to one of several competing formats. For the two most widely adopted formats, Eurorack (sometimes called 3U in reference to the height of modules, or Doepfer A-100 after the first branded system to adopt this panel height) and MOTM/MU (with a significantly larger 5U panel height), hundreds of competing companies produce modules for what are, in effect, open, nonproprietary standards. A “module” is a circuit that does one or a small number of operations and is attached to a front panel containing jacks and often user-controllable interfacial elements (e.g., knobs, sliders, buttons, toggles, capacitive touch plates). In order to make an “instrument,” users purchase multiple modules and then use patch cables to route voltages—both within and between modules. All contemporary modular synthesizers I have seen in performance contexts have contained between a dozen and several hundred modules. Whereas “fixed-architecture” (or all-in-one) synthesizers under the hood contain many circuits and hardwired connections that feel like a coherent singular instrument that reliably works the moment it is turned on, a contemporary modular synthesizer does nothing when initially plugged in. Moreover, a modular synth can easily be reconfigured if the user replaces any number of the modules within their “rack.” This makes the user process of “designing” an instrument an important site of creative play, but the inherent instability and changeability adds to the complexity of analyzing modular synthesizers *as instruments*.

Modular synthesizers provide a rewarding case study both for music studies and for STS for a number of reasons. As I hinted earlier, the expansion of interest in hardware modular synthesizers was an unforeseen outcome that continues to puzzle makers and users alike. They are expensive and bulky, hard to use, produce no sound when you first plug them in, and often sound horrible after you do (within the milieu, the terms *bloops and bleeps* and *robot farts* playfully address this). They do not define the sound of any chart-topping popular music genres, the market availability of any particular module is uncertain, and despite adhering to a very loose standard the modules often don’t work optimally with other modules without additional modules inserted in between. Even then, they may be temperamental. In other words, they fail to meet most of the conditions that Pinch (2008) described as necessary for

the so-called “success” of an instrument. Twenty-four years later there still is no “closure” with regard to the objects and they lack interfacial elements that Pinch argued were essential for widespread adoption (e.g., a piano-style keyboard). Contemporary modular synths perform primarily obscure/niche repertoires, and some of the most respected designers are outside of the public limelight and thus not “boundary shifters” in the manner of Bob Moog. Yet, as of August 2020, the Eurorack market alone (not including several other modular formats) consists of at least 7,777 unique modules produced by over 559 makers,¹ some of which have sold in quantities greater than the total number of original Minimoogs (12,000) that were ever made (Pinch and Trocco 2002: 214). Thus, contemporary modular synthesis presents an opportunity to reassess theoretical assumptions that are typically unquestioned about instruments, why people make and play them, and why they are (re)designed the way they are.

As is the case with other instruments that have been studied through critical organology (Bates 2012), modular synths are in large part constitutive of a culture, in this case a milieu culture (Webb 2007) that is most prominent in Europe, North America, Russia, Australia, and East Asia but with significant and growing local communities within South and Southeast Asia, North Africa, and Brazil. Webb’s milieu cultures consist of social milieus that are networked fields of cultural production, where the circulation and use of technologies and technological practices is central to their constitution, more so than cultural geographic concepts of locality or cultural studies concerns with resistance to a mainstream. In contrast to existing studies—both of synthesis and about regional instrument-specific cultural formations—I argue that we can only understand modular synthesis milieu cultures by attending to questions of *interface*, since questions about interface define the technical objects *and* create a problem space that ends up bringing together makers, users, and the numerous kinds of mediating roles (marketing, distribution, pedagogical, and user-generated content other than musical performance). Correspondingly, by doing so we expose some of the insufficiencies of extant theorization within both STS and music studies, neither of which has either a sufficiently coherent concept of interface or a conceptual framework for understanding instrumentality (Hardjowirogo 2017) in relation to questions of interface. More specifically, by looking at the varieties of entities that can be interfaced and the multisensory and affective qualities of interfacing, we become better equipped to tease apart distinctions made within milieu cultures between concepts of technological/user control and agency.

Hardjowirogo, writing within the context of a compendium of essays originating in the New Instruments for Musical Expression (NiME) conference series, provides a seven-part assessment of instrumentality that I find to be a constructive starting point. As she notes, the instrumental potential of an object can be measured via its:

- 1) sound production,
- 2) intention/purpose,
- 3) learnability/virtuosity,
- 4) playability/control/immediacy/agency/interaction,
- 5) expressivity/effort/corporeality,
- 6) immaterial features/cultural embeddedness, and
- 7) audience perception/liveness.

(Hardjowirogo 2017: 17–21)

I will not programmatically analyze all seven of these in relation to modular synthesis, but instead will focus predominantly on aspects four through seven. What is underdiscussed by Hardjowirogo, however, is what constitutes concepts such as “agency” or “interaction” or “corporeality”—or precisely where or how these might be experienced. Modular synthesizers provide an ideal case study to analyze just this, particularly with regard to the complexity of finding, sensing, analyzing, or theorizing the interface.

This essay will first briefly situate post-1995 modular synth milieu culture within a longer history of hardware modular synthesizers, focusing particularly on distinctive sociotechnical developments within the late 1990s online synthesizer design communities. Then, I will analyze six interfacial modes found within hardware modular synthesizers, especially those pertaining to the widely adopted Eurorack modular synth format, and contrast this ground-up and heterogeneous conceptualization of interface with extant reductive definitions within STS and media studies. The following section will show how some of the problems with instrumental interfaces have less to do with *what* is being interfaced than with the nuances of the look, feel, haptic response, and staging of interfacial elements. In

the last section I will consider issues of control and agency within modular synthesizer performances in order to show how a performative understanding of interfacial modes and interfacial aesthetics that is shared between performers and audiences becomes more artistically meaningful than any resultant musical “works.”

The research for this essay and my broader project concerning modular synthesizer milieu cultures results from a variety of qualitative source types and analytical methods. Background information about early developments in post-1995 synthesizer design came partly from trawling through the post history of the synth-diy mailing list, which I compared with similar discussions on contemporary message forums such as Muff Wiggler, Gearslut, and MODULARGrid. Through attending trade shows and modular meetups in the United States, the United Kingdom, and Germany, especially the annual Superbooth tradeshow/festival in Berlin, I learned more about how designers demonstrate prototype modules to prospective users and often subsequently redesign aspects of their module interfaces. The Eurorack milieu culture is notable for its passion for self-documentation, making the considerable body of interviews, podcasts, and product demos, especially Darwin Grosse’s “Art + Music + Technology” podcast and DivKid’s demo videos, excellent resources for analysis. When extant material lacked sufficient discussion of topics related to my research questions, I conducted formal and informal interviews with synth designers. Finally, I analyzed performance videos to understand the range of interfacial practices of amateur and professional synthesists in different countries working within a variety of genres.

Modular histories and the interface problem

Commercially available modular synthesizers from the 1960s and 1970s, initially made by Moog and Buchla but later by Arp and Serge (in the US), Korg and Roland (Japan), and EMS (UK), were expensive and bulky proprietary systems. “Modular” initially meant you chose a set of modules (discrete physical objects) from a small offering provided by a single manufacturer, but later, with the ARP2600 and Korg MS/PS-series synths, modular came to mean a workflow for synthesis where users connected parts of a monolithic instrument’s circuitry with patch cables. The main customers for Moogs, Buchlas, and ARPs were university electroacoustic music studios and successful popular/film music producers and composers (Figure 8.1). Moog was the first of these companies to experiment with abandoning modularity and producing a lower-cost, more portable, prerouted synthesizer, and their MiniMoog established an archetypal form and front panel look that continues to be copied by other manufacturers (Pinch and Trocco 2002: 224). It’s commonly but erroneously said that modular synthesizers disappeared from the market when affordable digital synthesizers became popular in the early 1980s. While Moog and Buchla temporarily stopped producing new modular/patchable synthesizer products in 1981–82, PAiA continued to offer smaller and more affordable DIY modular synths until the late 1980s, the EMS Synthesi and VCS3s have been made continuously since the 1960s, and Serge system synthesizers have been made in various forms since 1974 (Vail 2014). However, they weren’t sold in stores, and late 1980s to early 1990s popular music was dominated by mass-market digital synths and sample-based music production.



Figure. 8.1 The Moog Modular Opus 1 at the University of Michigan. Made in 1964, this was the first commercially sold Moog Modular, and originally owned by dance choreographer Alwin Nikolais. Photo by the author

By the mid-1990s, topic-focused electronic mailing lists were a sufficiently widely adopted medium that they became primary sites for nascent milieu cultures. The year 1995 saw a substantial increase in activity at the synth-diy and the Analogue Heaven mailing lists. It turned out that lots of people who had never met each other had some form of shared interest in analog synthesis; many had encountered these objects gathering dust at university composition studios, had made their own PAiA kit synthesizers, or loved an album that had been made with a particular modular synthesizer and were unable to replicate that sound with 1990s synthesizer technology. Others had dabbled with circuit-bending, which since Reed Ghazala's 1960s experiments with repurposing electronics for musical purposes had been a vibrant milieu culture that hardware-hacked everything ranging from Speak and Spell toys to Furbies into sound-producing devices (Ghazala 2005; Collins 2009; Rodgers 2010: 147–48; Pinch 2016). Since few of the participants in this milieu culture-in-formation were formally trained analog circuit designers, much of the first shared information included scans of circuit diagrams from 1970s electronics magazines including *Practical Electronics* and *Electronotes* (sharing was done both with and without permission).

None of the drivers of the modular synth revival was an established synthesizer company, and few participants at the time showed the intent to establish for-profit businesses. An exception was Dieter Doepfer, an occasional synth-diy participant who had run a successful business making controller keyboards for the German market prior to demonstrating his A-100 system at the 1995 Frankfurt Musikmesse trade show. In general, though, without the pressure of maintaining product lines that needed to sell in massive quantities, designers had relative freedom to experiment with both circuits and interfaces. Especially on synth-diy in the late 1990s, we see that this experimentation is also not wholly concerned with the *goals* or outcomes—products for a marketplace, or musical performance—but on the *process* and *experience* of designing and making synthesis modules. Hence, when Paul Schreiber introduced the MOTM format in 1998, the modules were sold in kit form, designed to be soldered together and possibly modded by prospective end-users. Soldering components to circuit boards is a multisensory experience often described as “therapeutic,” characterized by an “inner labor” the solderers go through to calm their nerves enough so that their hands can do the precise work required, by the haptic feedback transmitted through the handle of the soldering iron that indicates that solder is indeed flowing, by the smell of rosin flux fumes, and by the visual experience of components and circuit traces. By the late 1990s there were multiple competing formats, but all of the producers faced similar problems concerning economies of scale and manufacturing. You needed to make more than 10,000 of a gizmo

at a time to consider mass-manufacturing in China, and modular synthesis had only several hundred known participants. Thus, just as in the 1960s, objects were typically hand-soldered, with handmade faceplates. No one at the time had any sense that this might later become a financially profitable musical instrument sector.

Although Doepfer's synths had modest sales, as early as 1999 and for the next 15 years, Dieter Doepfer opined in most of his trade-show interviews that he feared the market for modular synthesis would disappear due to the high cost and physical bulk, in comparison with softsynths, which continued to improve in processing power, sound quality, and affordability. With the benefit of hindsight, I would argue the opposite: affordable softsynths—Propellerhead's ReBirth and Reason, Native Instruments Absynth and Reaktor, Spectrasonics Omnisphere, among many examples—created the bedrock of demand for hardware modular synthesis. Products such as these expanded the field of musicians who were attracted by the possibilities of allegedly limitless sound design and creative production, but the unwieldy user interfaces of softsynths, which depended almost exclusively on user mouse clicking on skeuomorphic visual representations of hardware gear alongside complex “menu diving” scenarios, made for an extremely frustrating and kinesthetically/haptically nonmusical user experience.

Even when software synthesis companies began developing tactile interfaces for softsynths they were greatly limited by the (effectively) 7-bit MIDI standard. A MIDI controller switches between up to 128 discrete steps rather than instigating a continuous and smooth change in that parameter, as is the case with analog gear, meaning that certain operations and basic audible aesthetic elements like continuous filter sweeps can't be achieved easily with common computer-hosted software. Equally significantly, the growth in computer-dependent professions and the use of computers for social media left computers as ambiguous objects for some synthesizer aficionados vis-à-vis their capacity for inspiring “creativity.” Ryan Gaston (2016) argues that the “multimodality” of computers is the key problem, since they are unlike acoustic instruments “where the user interface does not change and has only one set of intended uses. The synthesizer's relative inflexibility is actually what makes it function similarly enough to acoustic instruments to remain a relatable platform for collaboration with other instrumentalists.” Richard Nicol of Pittsburgh Modular notes: “I think it started as antilaptop. So many people were playing music on their laptop hunched over at shows. It's uninteresting, it's boring, there's no interaction in there...You don't have relationships with software, but you can have a relationship with a synthesizer.”² Christina Sealey of the Canadian dark techno duo Orphx mentions that her transition from laptops to Eurorack came from “being frustrated by being in front of a computer screen and finding that really unsatisfying. I'm a very hands-on, tactile person, and to feel stuck at the computer, even with a controller to operate it, it feels really limiting and I wanted something that was a bit more raw to use and had different possibilities for the sound.”³ Nicol and Sealey are not specifically opposed to synthesizers that are in part digital; ever since Scott Jaeger introduced his Harvestman line of digital modules in 2007, modular synthesis has been a significant milieu for experimentation with computation. The problem, instead, is primarily one of interface.

Theorizing interfaces

So far, I have been able to discuss some basic factors concerning modular synthesis without providing a clear definition of interface. That's not altogether surprising, since interface isn't something that has consistent definitions in music studies or in STS for that matter, even as the term is tasked with doing a lot of work. The range of things called interfaces in *Analog Days* include knobs and wires, a (computer) keyboard and mouse, a ribbon controller, the MIDI (musical instrument digital interface) standard, but most of all the piano-style keyboard eventually adopted by Moog (Pinch and Trocco 2002). An eclectic list! In his electronic dance music production research, Butler initially defines interfaces as “the actual site of mediation as materialized in a hardware or software surface” (2014: 7), but subsequently drops the mediation concept and reverts to using interface to mean material control elements. Things are much muddier in Ihde's (1991) phenomenological ruminations on the “interface” between science and technology; we never learn if that something is material, perceptual, human, mediating, conceptual, or some combination. In critical theory and media studies, we find irreconcilable approaches: Galloway notes that “interfaces are not simply objects or boundary points. They are autonomous zones of activity. Interfaces are not things, but rather processes that effect a result of whatever kind” (2012: vii), which would preclude most of Pinch's and Butler's thingly entities. In contrast, Bratton ruminates that interfaces are “any point of contact between two complex systems that governs the conditions of exchange between those systems” (2015: 220). This is surprising considering Bratton's customary indebtedness to Virilio, for whom interfaces are:

that drastically new surface that annuls the classical separation of position, of instant or object, as well as the traditional partitioning of space into physical dimensions, in favor of an almost instantaneous configuration in which the observer and the observed are roughly linked, confused, and chained by an encoded language from which emerges the ambiguity of interpretation of represented form-images, an ambiguity that returns to that of audio-visual media.

(Virilio 1991:52)

So interfaces could be parts of a more complex *material* technological object (Pinch and Trocco), a *site of mediation* (Butler), an unknowable “something” between two epistemes (Ihde), *points* that govern exchange between discrete objects (Bratton), *processes* and *not* points or things (Galloway), and the *dissolution* of subject and object and of the dimensionality of space itself (Virilio), all of which ignore the interfacial *surfaces* of chemistry. When considering any one interface, *at least* two of these *have* to be wrong; from even the most elementary understanding of mathematical topology, no concept/thing can be simultaneously a material object, a site, a surface, a point, a process that is not a point, *and* a dissolution.

Instead, I propose that we take seriously the multiple modes of interfacing that can happen during modular synthesizer “use.” Let us recall that synthesizers are notable for their morphological variety, and the primary problem space that brought synthesists back to hardware modular, more than questions of sound or creativity or novelty, was the perceived insufficiency of extant synthesizer interfaces (Scott 2016). For some users, the problem was how the tactile experience and look of synthesizers limited the emotional-affective potential of synthesizer encounters (Rodgers 2010)—especially the queering potential of what Roshanak Kheshti terms “synthgender” (Kheshti 2019: 32). For others problems lay in the disconnect between the locked-in feel of black-boxed synthesizer objects and the interest many synthesists had in actively designing or remaking them (Pinch 2016), or the inability to connect spaces of operations or activity together to create a complex instrument (exemplified by but not limited to the considerable problems with MIDI, see Lanier 2010 and Théberge 1997). Hence, by attending to the fullest interfacial variety within a space of practice (synthesis) that is largely defined in opposition to morphological alternatives (everything that isn’t hardware modular), we learn not just about the practice of synthesis, but about how technical individuals (Simondon 2017) cohere into technical ensembles that are regarded as musical instruments. Moreover, many of the distinctive features in the social life of synthesizers (Bates 2012)—from the “musical toys for boys” social formation that has restricted female participation in music technology in Britain (Born and Devine 2015) to the queering effects of synthesizer use—likely have more to do with the lived experience of materiality that cuts across the design/making and “use” of interfaces than they pertain to sound or music.

Shared within contemporary hardware modular synthesizers is the sense that “everything has to feel analog” (Botstein 2014), as put by Tony Rolando, the lead designer of the acclaimed Make Noise modules, by which he means (in part) the use of interfacial elements such as knobs, sliders, buttons, jacks, capacitive touch, and force-sensing resistors, and minimization or elimination of menu-diving graphical interfaces. Unlike earlier periods in music technological development where the so-called “analog” and “digital” instruments often had quite divergent interfacial elements, aesthetics, and feel, this is not typically the case with contemporary modules, the most popular of which feature a “knob-per-function” interface. Therefore no *a priori* analog/digital distinction should be assumed—either regarding sound or interface. For Scott Jaeger, designer of the Industrial Music Electronics and Harvestman modules, many of which are based on code he’s written that runs on a PIC32 chip (supplemented by analog circuitry), “‘digital’ doesn’t mean a single data slider and contempt for your market. It’s more about the possibilities of strange and interesting sounds that can be reached by computation, nothing to do with the user interface at all.”⁴

From my analysis of performances, online user discussions about patching and specific modules, and interviews with designers (some that I’ve conducted and others produced within Eurorack milieu cultures), I’ve determined that there are at least six modes of interfaciality that might be at play. While some modules are locked in (Lanier 2010) or “scripted” (Akrich 1992) to one interfacial mode, others vary depending upon the expertise of the user or the way that they incorporate it into a patch. First, modules may appear to users as black-boxed entities (Shindell 2019) with a limited number of interfacial parameters and one or two variable controls at most that become the site of performative play. For example, some modules on the market essentially facilitate a single kind of operation or means of changing sound, whether a spring reverb or a distortion unit where the user mainly controls the amount of distortion but is locked in to the circuit and component choices made by the designer. Quite a few feature-limited oscillators work this way—they produce a predictable analog waveform at a frequency relative to the voltage they receive. More complex

objects, though, which nominally expose more of the circuit to the user, may be black-boxed in practice, especially if some combination of settings enables users to achieve a desired sound specific to a musical genre. For example, I've frequently seen prospective consumers who want an 808-style kick sound to make techno, a 303-style synth voice to make acid leads for trance, or the swirly granular ambient reverb that the Mutable Instruments Clouds module brings to generative or "houseplant" ambient. None of these depends upon real-time tactile control. We could term this motivation *interfacing to a genre*, considering not just the genre's audible properties, but also its social scene dimensions. This mode of interfaciality is routinely made social when the numerous amateur YouTube videos that show users "how to" replicate generic conventions become the fulcrum of online social interaction. However, within a large subset of the modular synthesis milieu cultures, these motivations may be perceived as artistically uninspiring, and not in keeping with the "modular spirit" (Bates 2021). Especially on online forums, users desiring to do this are often advised by regulars to purchase much more affordable fixed-architecture synths or software synths rather than expensive modular setups.

Second, we find interfaces that provide a less mediated, more direct connection to an electrical property or operation specific to one or more electrical components within the object. This is a very common basis for engineers to design new modules. For example, Joran van Gremberghe noted that many of his designs began by exploring the possibilities of Burr Brown op amps (interview, 9 May 2019), while William Mathewson detailed how his Synchrodyne filter module arose from "happy accidents" while playing around with a switch capacitor chip (interview, 27 October 2017). Nicolas Collins's textbook popularized the creative misuse of a 40106 "Schmitt-Trigger" CMOS logic chip as a dirty oscillator (Collins 2009), which inspired numerous spin-off designs. In analyzing the history of internet discussion of synthesizers, I noticed that in the mid-1990s comparatively few people (aside from amateur or professional synthesizer designers) were even aware what any of the specific parts were, unless a synthesizer they owned that depended upon a part stopped working and they had to search for a replacement part. By the early 2000s, internet discussion had had a major role in expanding the types of individuals involved in discussing once esoteric matters, although this does not mean that *all* parts are equally well known or discussed in this manner.

Components-as-interfaces may exhibit audible or visible properties if the electrical component in question is working with audio/video signals directly, but may not directly have a distinctive "sound" if the component is performing basic mathematical operations that are applied to control voltages (CVs) intended to control other Eurorack modules. In contrast to black-boxed interfaces, here we find two kinds of interfacing at play: the more obvious interface between user and component as presented via tactile controls and patch points, and a less obvious one, where within the designer's schematic, the component in question interfaces between other circuit-level components in an interesting way. This dual nature is key and becomes part of the justification for bringing this product to market. Such products succeed as products mainly because today there are prospective users who have a sufficient amount of electrical engineering knowledge *and* an interest in the idiosyncrasies of particular components.

Third, because post-1995 modular synthesis contains many modules that are in part digital, controls may provide access to digital objects, code, and digital signal processing (DSP) via a more "analog"-style interface (e.g., knobs and jacks rather than a computer mouse/keyboard/display). To support this, a number of microcontrollers, including the Teensy (which uses an ARM Cortex chip to run a fork of Arduino software called Teensyduino), the PIC (which can run either assembly or C-language code, see Lynch and Marchuk 2014), and the Spin FV-1 (which hosts code libraries for time-based effects like reverb and delay) are widely used to perform complex computations or digital signal processing (Popp and Soria-Luz 2015).⁵ Despite the presence of computation, the inputs and outputs of the module (both control voltages and audio) are all analog. Such modules, when used to produce or modify waveforms, may end up having specific "digital signatures" (Brøvig-Hanssen and Danielsen 2016), such as the aforementioned Mutable Instruments Clouds module when set to function as a reverb. But many others do not deal with sound directly. For example, the XAOC Devices Zadar produces complex envelopes—continuous patterns of rising and falling voltages—which can be used to "shape" the pitch, amplitude, timbral quality, or any other aspect of a sound made or manipulated in another module—but the Zadar doesn't typically have a sound of its own.

Fourth, while some computation modules are black-boxed in the sense that users cannot alter the underlying code, others provide hackable firmwares that allow interested users to modify the code or write their own. Here, the interface in question is not the front panel at all, but instead the integrated development environment (IDE), which provides a set of standard code/class libraries and a framework for users to create code objects. After changing the code and reinstalling the module in the user's rack, these modules then revert to the third kind of interfaciality.

Fifth, a number of “utility” modules lack direct user control (i.e., they only have patch point jacks and no tactile controls) but are vital since they enable modules to interface with other modules. This includes things like MIDI to CV converters, external interface modules that scale the Eurorack audio levels to the lower levels of studio gear or guitar pedals, buffered multiples that produce multiple identical copies of an incoming signal, or unity gain mixers which add together control voltages or audio sources. It also includes some staples of analog synthesis that do change the sound of audio, such as waveform rectifiers (which take a normal incoming waveform that oscillates between positive and negative voltages and “flip” all the negative voltages to positive ones, creating an interesting kind of distortion), and many kinds of logical operators (e.g., performing an “exclusive OR” mathematical operation between two voltages, which will output a voltage only if one voltage is positive and the other zero). The lack of human-object interfaciality might make these “utility modules” initially seem less interesting. But as I have seen routinely discussed online and at trade shows, experienced users tend to have a lot of such modules and consider them essential for making an effective patch, in part since the major challenge of modular synthesis is getting a variety of modules from different manufacturers to operate with each other in a desired way, and in part since basic mathematical operations applied to voltages ($A \times B$, $A \text{ XOR } B$) sometimes contribute to the most interesting sonic outcomes. Put another way, while modular synthesizer users do want to create a fun and playable device with a tactile interface with an inspiring analog feel (Bates 2021), they can’t do so without overcoming the quirks and interoperability concerns of the fun modules they do have—and without patching modules that can do some math for them.

Sixth, many modules employ force-sensing resistors or capacitive touch plates that respond to the user’s body, while others (for example, the Schlappi 100 Grit distortion module) expose certain parts of the circuit traces to the user in the form of conductive metal parts. These make the user’s body part of the circuit and signal flow itself, where the body acts like an unpredictable resistor or as a patch cable with “dirty” capacitance properties. In other words, for these modules the human body itself is the interface between modules. Even when the users are not actively electrocuting themselves or contributing their electrical discharges, in a more general sense the entanglement between synthesizers and the bodies/identities of synthesists has been long discussed, for example with regard to Suzanne Ciani’s conflicted “analog human-machine identity” with her Buchla synth (Pinch and Trocco 2002: 170) or Wendy Carlos’s concern for the “human touch” potential of Moogs (Kheshti 2019: 58).

Particular modules may or may not be designed to facilitate one of these interfacial modes, and in some cases modules designed to be more interactive end up being used in black-boxed ways. Some modules present a degree of “interpretive flexibility” (Pinch 2008: 471), meaning that how and what they interface is contingent upon the choices and competence of the user. However, the fifth and sixth modes do not have this interpretive flexibility; as transductive technologies (Helmreich 2015), their operation is not primarily dependent upon user attitudes.

This sixfold nature of interface (Figure 8.2) led to the emergence of a vibrant manufacturer/designer ecosystem, and a proliferation of numerous variants of each technological object type. For example, there are now hundreds of “filters” on the market. They differ with regard to the kind of circuit used to realize them (e.g., state-variable, ladder), with regard to the specific components that are used to realize that circuit (passive filters, vactrols, transistor arrays, integrated circuits, digital algorithms), and with regard to accommodating different ways that users might choose to use an object. This ranges from “set it and forget it” black-boxed filter modules (sound in, filtered sound out), to filters controlled by CVs produced or altered by other modules (an example of object-object interfacing), to filters providing users with intermittent or ongoing tactile control of parameters. But “filters” are no longer just for changing the sound or visual appearance of a signal, and many are designed explicitly to be “misused.” When pushed to self-oscillation, filters become full-fledged sound/video generators (Välimäki and Huovilainen 2006). Accordingly, users might experientially learn that oscillation is a property of electricity that can happen anywhere (not just in objects labeled “oscillators”) and can be leveraged for numerous tasks. Cumulatively, the tactile, experiential, multisensory engagement with modules, whether or not accompanied by textbook knowledge or prior design experience, provides the user with more capacity to understand synthesizer modules *and* their key components as technical individuals (Simondon 2017), and to perceive the potential for instrumentality (Hardjowirogo 2017) within aggregate ensembles.

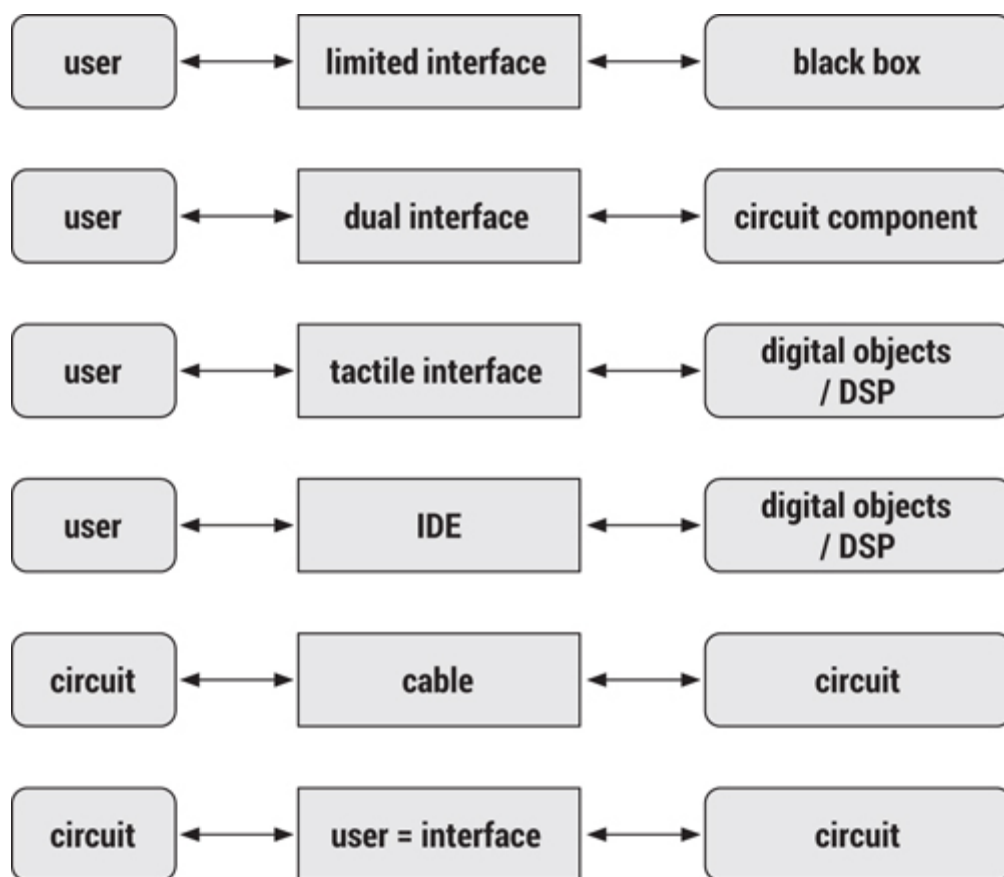


Figure 8.2 Simplification of the six interfacial modes from the material (user/module) perspective

Staging and experiencing the knob

Multifaceted and multisensory experiences of interfaciality (Pink et al. 2016), resulting from interfacial modes that increase *and* decrease user perceptions of control and agency (Bijsterveld and Schulp 2004; Navs 2016), are specifically what allow users, makers, and audiences to perceive modular synthesizer technical ensembles (Simondon 2017) *as* instruments. However, it is not simply a matter of the particular interfacial mode being performed, but the aesthetic and ludic aspects of the multi-interfacial performance. Here I'm reminded of Kodwo Eshun's description of the synthesizers of the great funk musician Bernie Worrell as being “psychokinaesthetic” (Eshun 1998: 149).

While a knob does present an opportunity for turning, not just any knob will do, and not all knob turns result from prior design decisions that put the knob there in order to increase the user's sense of control. Whereas designers like Paul Schreiber note that the high cost of Synthesis Technologies' modules comes from custom potentiometers rated for “two million turns,”⁶ and Scott Jaeger is so picky about the potentiometers and knob coverings for his Industrial Music Electronics modules that he even specified the kind of grease and knob ballistics to the component manufacturer,⁷ there are no precision alternatives to the widely derided “miniature” knobs that are very difficult to turn but allow many more interfacial elements to be crammed into a small space. Therefore, knobs are a class of object that can produce a range of affective experiences and performative outcomes. This is compounded by the staging of the knobs on the front panel, where some designers label the knobs with *technical* descriptors (e.g., frequency coarse, FM level), others with *poetic/catachrestic* captions (e.g., the knobs labeled “furthrrrrrr” and “mood index” on the endorphin.es Shuttle System), or others yet with abstract and artistic *icons* lacking text (e.g., many controls on devices by the companies Make Noise, Alright Devices, and Whimsical Raps).

Interfacial modes, control element design, and element staging demonstrate that the popular *dictum* “knobs afford turning” (Norman 2013: 13) and, for that matter, the “affordance” concept this is derived from (Gibson 1979) have only limited utility for analyzing technological interfaces. Not all knobs are turned, and even for those that are, not all afford turning for the same reasons or in the same way. Knob turning that is intended to change frequency, a mood index, a diagonal line, or the distorted luminosity of a lime green dinosaur are *experientially* different from each other

because the catachrestic information works together with the haptic response and change in the sound coming out of the instrument to produce qualitatively different sensations in the user's body. Specialized concepts like knob ballistics or turns ratings wouldn't have become part of the modular synthesis milieu culture vocabulary if the longstanding problematization of interfaces hadn't been so central to the formation of the milieu culture to begin with—or in other words, if all knobs had indeed afforded turning.

Control, agency, and instrumentality

For any object (synthesizers or otherwise) to be regarded as “a musical instrument,” it has to demonstrate its capability to produce some form of sound that is perceived (by at least someone) to be conducive to some form of musical expression. But many kinds of objects have the capacity to do this, and this by itself doesn't really tell us anything of use, since some of the Eurorack modules I've discussed in this essay are available as free or low-cost plugins for the free software VCV Rack where they nominally sound *identical* to the hardware version they were based off. The difference, then, is between a tactile object that provides a sufficient quality of haptic feedback and a skeuomorphic on-screen representation of the same (Gross et al. 2014). Users wouldn't choose the hardware rather than the software due to the degree of parameterized “control” provided: the same number and kind of control elements can be skeuomorphically represented. Instead, hardware modular performers talk often about “happy accidents” and their frequent *lack* of control over the instrument when reacting to the tactile and haptic experience of patching and knob turning.

For many participants in the modular synthesis milieu culture, it matters most *how* a particular performance was realized. The audience for events often consists largely of other synthesists, and much of the event may be devoted to artists describing their instrument and the patch they performed (Figure 8.3). While there is no one obligatory embodied performative norm, and generative pieces that unfold with no direct performer intervention are not necessarily valued higher or lower than ones where the performer was frantically turning knobs and pushing sliders, within many of the local or platform-specific subsets of the modular synthesis milieu culture it is more frowned upon to use piano-style keyboards, prerecorded backing tracks, or preset rhythms from external (nonmodular) drum machines. In other words, the “composition,” if we want to persist in uttering a term like that, is in the user design of the instrument and the patch diagram that the performer devises (Magnusson 2019: 118; Navs 2016)—which *always* results from a tactile and haptic experience. Not surprisingly, audiovisual media including live performances, “making of” videos, and module use tutorials have been much more important for disseminating modular synthesis praxis than audio-only albums. Audiences want to see how a performance was made, and they positively respond to audiovisual evidence that the “performer” struggled with the instrument to make that performance happen (before and/or during the performance). This translates (Latour 1987) immediately when a three-dimensional hardware object with patch cables and knobs is an active participant, but not when skeuomorphic representations and software objects are involved. In part, this is because it's hard to tell if a software patch resulted from user labor or instead was a preset or prepackaged soundbank, as was commonly the case with performances of 1970s to 1990s digital instruments such as the Yamaha DX-7 or Fairlight CMI that had notoriously difficult or unpleasant interfaces (Théberge 1997: 75–77; Harkins 2019). At least within this electronic music milieu culture, musical sound or “works” have much less of a role in constituting the milieu culture than does the sharing of staged encounters with material technological objects, especially those that provide evidence of human effort and labor expended in the service of the instrument.

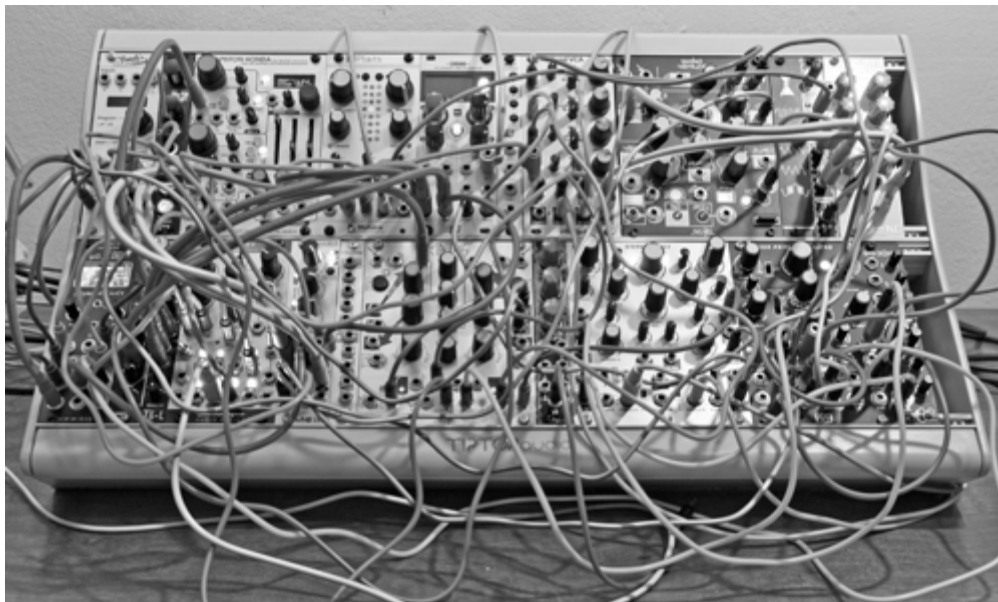


Figure. 8.3 A simple performance patch of the author's Eurorack-format modular synthesizer. Photo by the author

Richard Devine, perhaps the best-known performer, collector, and beta-tester of modular synthesizers, routinely mentions that when he unplugs a patch he'll never be able to produce that same work or same sound again. The inability to precisely replicate works is not a problem. Hardware modular synthesis, despite fetishizing "controllable" modules, is not about users projecting the illusion of control over their instrument, even as the acquisition of more modules provides users with more things with which to tactilely engage. Instead, "it's all about the interface,"⁸ or more precisely, about the numerous modes of interfacing that I have explored, the multisensory and affective aspects of interfacing, and the social benefits accruing from the mediatization and sharing of interfacial encounters.

Acknowledgments

Aspects of research related to this chapter were presented at the Society for the Social Studies of Science annual conference (2017), UC Berkeley (2017), the University of Michigan (2018), Australian National University (2018), Stony Brook (2019), Wesleyan University (2020), and at the Social Distinction in the 21st Recording Studio symposium at the University of Lethbridge. I thank Jocelyne Guilbault, Lester Monts, Samantha Bennett, Margarethe Adams, Jane Alden, and Amandine Pras for the invitations and for their critical engagement with the topics, and the attendees for their questions and critiques. I also thank Joran van Gremberghe, Scott Jaeger, Eric Schlappi, and William Mathewson for their willingness to participate in interviews, and Christopher Haworth and Ladi Dell'aira for their attentive reading and comments on an earlier version of the manuscript. All mistakes remain my own.

Notes

- 1 Based on data aggregated at <https://www.modulargrid.net/> [access date: 29 August 2020].
- 2 Meet the Makers: Richard Nicol (Pittsburgh) (2015). <https://www.youtube.com/watch?v=kG8PN4CAG1I> [access date 30 August 2020].
- 3 *I Dream of Wires*: Orphx Extended Interview (2015). <https://www.youtube.com/watch?v=pdPvK5-k088> [access date 30 August 2020].
- 4 Paul Clark, "Scott Jaeger Interview" (2010). <http://www.electronicmusic.com/features/interview/scottjaeger.html> [access date: 30 August 2020].
- 5 This should not be conflated with the concept of interface classes in object-oriented programming languages, or application programming interfaces (APIs), which allow external computers and code to access a program running on a computer.
- 6 Ken Flux Pierce, "The SynthSummitShow: Paul Schreiber Synthesis Technology E520" (2019). https://www.youtube.com/watch?v=i_ihwxZB5KY [access date 10 November 2019].
- 7 Scott Jaeger, interview, 10 May 2019.
- 8 William Mathewson, interview, 27 October 2017.

References

- Akrich, Madeleine. 1992. "The De-Description of Technical Objects." In *Shaping Technology/Building Society: Studies in Sociotechnical Change*, edited by Wiebe E. Bijker and John Law, 205–224. Cambridge, MA: MIT Press.

- Bates, Eliot. 2012. "The Social Life of Musical Instruments." *Ethnomusicology* 56 (3): 363–395.
- . 2021. "Feeling Analog: Using Modular Synthesizers, Designing Synthesis Communities." In *Shaping Sound and Society: The Cultural Study of Musical Instruments*, edited by Steven Cottrell. New York: Routledge.
- Bijsterveld, Karin, and Marten Schulp. 2004. "Breaking into a World of Perfection: Innovation in Today's Classical Musical Instruments." *Social Studies of Science* 34 (5): 649–674.
- Bonanno, Carmine. 1984. Guitar Controller for a Music Synthesizer. United States patent US 4,630,520A, filed November 8.
- Born, Georgina, and Kyle Devine. 2015. "Music Technology, Gender, and Class: Digitization, Educational and Social Change in Britain." *Twentieth-Century Music* 12 (2): 135–172.
- Botstein, Sam. 2014. "Tony Rolando (Make Noise)." *The Distillery*. September 24, 2014. <https://www.stitcher.com/podcast/sam-botstein-2/the-distillery/e/36619312>.
- Bratton, Benjamin H. 2015. *The Stack: On Software and Sovereignty*. Cambridge, MA: MIT Press.
- Brøvig-Hanssen, Ragnhild, and Anne Danielsen. 2016. *Digital Signatures: The Impact of Digitization on Popular Music Sound*. Cambridge, MA: MIT Press.
- Butler, Mark J. 2014. *Playing with Something that Runs: Technology, Improvisation, and Composition in DJ and Laptop Performance*. New York: Oxford University Press.
- Collins, Nicolas. 2009. *Handmade Electronic Music: The Art of Hardware Hacking*. New York: Routledge.
- D'Errico, Mike. 2016. "Interface Aesthetics: Sound, Software, and the Ecology of Digital Audio Production." PhD diss., UCLA.
- Eshun, Kodwo. 1998. *More Brilliant than the Sun: Adventures in Sonic Fiction*. London: Quartet Books.
- Galloway, Alexander R. 2012. *The Interface Effect*. Cambridge, MA: Polity Press.
- Gaston, Ryan. 2016. "Plays Well with Others: Regarding Modular Synthesizer in Collaborative Performance Practice." *EContact!* 17 (4). https://econtact.ca/17_4/gaston_playswell.html.
- Ghazala, Reed. 2005. *Circuit-Bending: Build Your Own Alien Instruments*. New York: Wiley.
- Gibson, J. J. 1979. *The Ecological Approach to Visual Perception*. Boston, MA: Houghton Mifflin.
- Gross, Shad, Jeffrey Bardzell, and Shaowen Bardzell. 2014. "Skeu the Evolution: Skeuomorphs, Style, and the Material of Tangible Interactions." *8th International Conference on Tangible, Embedded and Embodied Interaction*. Munich, DE.
- Hardjowirogo, Sarah-Indriyati. 2017. "Instrumentality: On the Construction of Instrumental Identity." In *Musical Instruments in the 21st Century: Identities, Configurations, Practices*, edited by Till Bovermann, Alberto de Campo, Hauke Egermann, Sarah-Indriyati Hardjowirogo, and Stefan Weinzierl, 9–24. Singapore: Springer.
- Harkins, Paul. 2019. *Digital Sampling: The Design and Use of Music Technologies*. Boca Raton, FL: CRC Press.
- Helmreich, Stefan. 2015. "Transduction." In *Keywords in Sound*, edited by David Novak and Matt Sakakeeny, 222–231. Durham, NC: Duke University Press.
- Ihde, Don. 1991. *Instrumental Realism: The Interface between Philosophy of Science and Philosophy of Technology*. Bloomington, IN: Indiana University Press.
- Kheshti, Roshanak. 2019. *Wendy Carlos's Switched-On Bach* (33 1/3). New York: Bloomsbury.
- Lanier, Jaron. 2010. *You Are not a Gadget: A Manifesto*. New York: Alfred A. Knopf.
- Latour, Bruno. 1987. *Science in Action: How to Follow Scientists and Engineers Through Society*. Cambridge, MA: Harvard University Press.
- Lynch, Kevin M., and Nicholas Marchuk. 2014. *Embedded Computing in C with the PIC32 Microcontroller*. Evanston, IL: Northwestern Mechatronics Design Lab.
- Magnusson, Thor. 2019. *Sonic Writing: Technologies of Material, Symbolic, and Signal Inscriptions*. New York: Bloomsbury Academic.
- Meintjes, Louise. 2003. *Sound of Africa!: Making Music Zulu in a South African Studio*. Durham, NC: Duke University Press.
- Navs. 2016. "Basic Electricity: An Appeal for a Greater Understanding of Rudimentary Modular Functions." *EContact!* 17 (4). https://econtact.ca/17_4/navs_basicelectricity.html.
- Norman, Don. 2013. *The Design of Everyday Things*. Rev. ed. New York: Basic Books.
- Oudshoorn, Nelly, and Trevor J. Pinch. 2003. *How Users Matter: The Co-Construction of Users and Technologies*. Cambridge, MA: MIT Press.
- Pinch, Trevor. 2008. "Technology and Institutions: Living in a Material World." *Theory and Society* 37 (5): 461–483.
- . 2016. "'Bring on Sector Two!': The Sounds of Bent and Broken Circuits." *Sound Studies* 2 (1): 36–51.
- Pinch, Trevor, and Karin Bijsterveld. 2004. "Sound Studies: New Technologies and Music." *Social Studies of Science* 34 (5): 635–648.
- Pinch, Trevor, and Frank Trocco. 2002. *Analog Days: The Invention and Impact of the Moog Synthesizer*. Cambridge, MA: Harvard University Press.
- Pink, Sarah, Heather Horst, John Postill, Larissa Hjorth, Tania Lewis, and Jo Tacchi. 2016. *Digital Ethnography Principles and Practice*. London: SAGE.
- Popp, Constantin, and R. Soria-Luz. 2015. "Developing Mixer-Style Controllers Based on Arduino/Teensy Microcontrollers." In *The 12th Sound and Music Computer Conference*, Maynooth University, Maynooth, Ireland, 37–41.
- Rasmussen, Anne K. 1996. "Theory and Practice at the 'Arabic Org': Digital Technology in Contemporary Arab Music Performance." *Popular Music* 15 (3): 345–365.
- Rodgers, Tara. 2010. *Pink Noises: Women on Electronic Music and Sound*. Durham, NC: Duke University Press.
- Scott, Richard. 2016. "Back to the Future: On Misunderstanding Modular Synthesizers." *EContact!* 17 (4). https://econtact.ca/17_4/scott_misunderstanding.html.
- Sen, Sourya. 2019. "Developing a Hybrid Video Synthesiser for Audiovisual Performance and Composition." MA thesis, Aalto University, Finland.
- Shindell, Matthew. 2019. "Outlining the Black Box: An Introduction to Four Papers." *Science, Technology, & Human Values* 45 (4): 567–574.
- Simondon, Gilbert. 2017. *On the Mode of Existence of Technical Objects*. Translated by Cécile Malaspina and John Rogove. Minneapolis, MN: Univocal Publishing.
- Théberge, Paul. 1997. *Any Sound You Can Imagine: Making Music/Consuming Technology*. Middletown, CT: Wesleyan University Press.
- Vail, Mark. 2014. *The Synthesizer: A Comprehensive Guide to Understanding, Programming, Playing, and Recording the Ultimate Electronic Music Instrument*. New York: Oxford University Press.
- Välimäki, Vesa, and Antti Huovilainen. 2006. "Oscillator and Filter Algorithms for Virtual Analog Synthesis." *Computer Music Journal* 30 (2): 19–31.
- Virilio, Paul. 1991. *The Lost Dimension*. Translated by Daniel Moshenberg. New York: Semiotext(e).
- Webb, Peter. 2007. *Exploring the Networked Worlds of Popular Music: Milieu Cultures*. New York: Routledge.